

When Do Graph Neural Networks Help in Cross-Sectional Stock Ranking?

A Multi-Seed, Multi-Universe, Cost-Aware Study of the U.S. S&P 500

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ABSTRACT

Graph neural networks (GNNs) are often reported to improve cross-sectional stock ranking by encoding relations among stocks as graph edges. Many reported gains, however, come from single-seed, single-split evaluations and do not adjust for multiple testing. We ask a narrower question: when, if at all, does the graph help? We evaluate an equal-budget tuned ladder from LightGBM to a heterogeneous relational GNN (L0–L7) over twelve expanding quarterly walk-forward folds and ten seeds on the U.S. S&P 500.

We pre-register two confirmatory families. The predictive family asks whether any tuned model beats tuned LightGBM. Hansen’s SPA test does not reject in either feature universe ($p = 0.277$ and $p = 0.077$), and the benchmark comparisons are underpowered, so the result is a non-rejection rather than evidence of equality. Local DM contrasts place the observable signal in the non-graph MLP and, most robustly, *against* the graph: adding a correlation-GAT to the MLP reduces IC ($\Delta\text{IC} = -0.012$), and this penalty is BH-significant in *both* feature universes and never changes sign when any single quarter is dropped. The MLP’s gain over LightGBM ($\Delta\text{IC} = +0.015$) is significant only in the leak-selected universe and falls below its detectable-effect threshold, so we read it as suggestive (Limitation L1).

The edge-attribution family freezes the tuned correlation-GAT operating point and varies only the edge set. None of six edge contrasts survives Benjamini–Hochberg control, and all six are underpowered. A transaction-cost crosswalk reinforces the same caution: the two load-bearing Universe-C ladder contrasts (MLP–LightGBM and correlation-GAT–MLP) survive 10 bps, whereas the tuned news-edge IC underperformance is near zero in net Sharpe, fragile across folds, and absent in the fixed-operating-point causal test. We therefore treat the evidence as conditional findings and failure modes, not as architecture rankings.

KEYWORDS

cross-sectional stock ranking, graph neural networks, walk-forward evaluation, Hansen SPA, multiple-comparison control, transaction costs

1 INTRODUCTION

Cross-sectional stock ranking asks whether a model can predict, each trading day, the relative ordering of next-period returns

across a fixed stock universe—here, the U.S. S&P 500. The economic object is a dollar-neutral long-short portfolio: correctly ranking the top and bottom deciles creates a tradable signal, whereas rankings close to chance are absorbed by transaction costs. We measure ranking quality with the *Information Coefficient* (IC), the daily cross-sectional Spearman rank correlation between predicted scores and realized forward returns.

Recent GNN papers on stock ranking, including Feng et al. [8], Kim et al. [14], and Sawhney et al. [21], report sizable IC gains from message passing over inter-stock relations such as sector membership, Wikidata links, and news co-mentions. Much of this evidence comes from one seed and one split. That design leaves three concerns.

The first is seed dispersion. Even after tuning and seed averaging, per-arm 95% bootstrap confidence intervals are often as wide as the IC point estimates. The Universe-C correlation-GAT, for example, has pooled IC 0.0224 with a 95% interval $[-0.0080, +0.0559]$; retuning does not remove this dispersion. Our confirmatory tests use the seed-mean signal as the estimand (§3.3), so this dispersion is a deployment precision caveat rather than part of family-wise inference.

The second concern is regime concentration: one high-dispersion quarter can dominate an annual estimate. The third is capacity confounding. A tuned edge ablation comparing differently sized networks mixes whether the edge helped with whether the larger network helped. A causal reading requires holding the operating point fixed.

We address these concerns with one tuned model ladder and two pre-registered confirmatory families. The first family is predictive. It asks whether any tuned arm improves on tuned LightGBM, using Hansen’s Superior Predictive Ability (SPA) test [11] over nine candidates. We use Diebold–Mariano / Harvey–Leybourne–Newbold (DM/HLN) pairwise tests [6, 12], with Benjamini–Hochberg (BH) false-discovery-rate control [2], as local evidence along the ladder. The second family is an edge-attribution family: it freezes the full hyperparameter vector of the correlation-graph operating point and changes only the edge set. The resulting matched ΔIC estimates the edge effect at L2’s tuned operating point; it is not an estimate of the best attainable performance of each edge set after retuning (§3.3).

Two feature universes. The two universes contain the same stocks and differ only in node features. **Universe B** is a

10-dimensional price-volume baseline. **Universe C** is a 51-dimensional subset of Qlib’s Alpha158 library [25], selected from the project’s earlier top-15 factor groups. All feature values are measured at $T - 1$. The comparison asks whether graph structure is more useful when node features are sparse or already rich.

Headline findings. No tuned arm is confirmed to outperform tuned LightGBM. Hansen’s SPA test does not reject in either universe (Universe B $p_{\text{consistent}} = 0.277$, Universe C $p_{\text{consistent}} = 0.077$), and the benchmark comparisons are underpowered because the observed gaps fall below the 80% minimum detectable effect. The result is therefore a non-rejection, not evidence that the models are equal.

The local ladder shows where any neural lift comes from, and its most robust signal is negative for the graph. Adding a correlation-GAT to the MLP’s features *reduces* IC (L2–L1): this contrast is BH-significant in *both* universes (Universe B $\Delta\text{IC} = -0.0133$; Universe C -0.0119 , $p = 6.9 \times 10^{-6}$), exceeds its 80% minimum detectable effect in Universe C, and does not change sign when any single test quarter is dropped (0/12 folds in both universes). Because it also holds in the clean price-volume Universe B, selection leakage cannot explain it. The positive direction is weaker: the tuned MLP beats LightGBM only in the leak-selected Universe C (L1–L0 $\Delta\text{IC} = +0.0148$, HLN $p = 0.011$, BH-reject; the magnitude-matched Universe-B contrast does not survive BH) and sits below its detectable-effect threshold, so we read it as suggestive (Limitation L1). These are local contrasts, not global SPA wins. The edge-attribution family finds 0/6 BH-FDR rejections, with 6/6 contrasts underpowered. At 10 bps, the two central Universe-C contrasts survive transaction costs, whereas the tuned news-edge underperformance does not carry to net Sharpe and is not reproduced by the fixed-operating-point causal test. The evidence is therefore a set of conditional findings and failure modes, not an architecture ranking.

2 RELATED WORK

GNN baselines for stock ranking. Several recent papers use graph structure to model cross-sectional stock predictability. Feng et al. [8] learn to rank over industry and Wikidata relation graphs. Kim et al. [14] introduce HATS, a hierarchical graph-attention network over 75 Wikidata relation types. HATS motivates our L7 rung, but L7 is only a HATS-style adaptation: it uses three relation types (correlation, GICS sector, and news co-occurrence) and a 21-day ranking head, so L7–L2 is subject to relation starvation.

Other architectures relax the graph more aggressively. Cheng and Li [5] infer dynamic firm relations through unmasked attention, and Li et al. [15] alternate intra- and inter-stock attention. Our L6 rung is a controlled proxy for this dense-attention family, but it does not reimplement AD-GAT’s dynamic-relation inference or MASTER’s market-gated attention. Related work also includes spatiotemporal hypergraphs [21], pattern-specific routing [16], MLP-only token mixing [7], and multi-relational dynamic graphs [20]. The common feature is richer relational structure evaluated without multi-seed, multiplicity-controlled, cost-aware protocols.

Methodology-oriented empirical finance. Our design also follows a methodological line in empirical asset pricing that treats multiple testing, dependence, and sample splitting as first-order concerns. Gu, Kelly, and Xiu [9] set the recursive out-of-sample benchmark for machine-learning return prediction and find no gain beyond shallow networks. Hou, Xue, and Zhang [13] replicate 452 cross-sectional anomalies under a common protocol: 65% fail even the single-test hurdle ($|t| \geq 1.96$), and the failure rate rises to 82% under their multiple-testing hurdle. Avramov, Cheng, and Metzger [1] show machine-learning long-short profits concentrate in hard-to-arbitrage stocks and shrink under value weighting and cost screens, motivating our gross-and-net cost layer. López de Prado’s [17] purge-and-embargo validation and block-bootstrap methods shape our walk-forward design. Hansen [11] motivates the SPA test; Politis and Romano [19] provide the stationary bootstrap, and Newey and West [18] provide the HAC standard errors.

Position relative to prior work. To the best of our knowledge, this is the first S&P 500 stock-ranking study to combine ten seeds, a 12-fold expanding walk-forward design, a tuned ladder, two pre-registered confirmatory families, gross and net portfolio evaluation, point-in-time news handling, and disclosure of a tuned-configuration stability failure. Closely related GNN-ranking papers typically report a single split and seed, without multiple-comparison control or a gross/net cost ladder; we treat those as reporting differences, not as evidence that earlier methods could not be evaluated under this protocol.

3 METHODOLOGY

3.1 Walk-forward cross-validation

Our validation design is a 12-fold *expanding-window walk-forward*. In fold k , training ends at a quarter boundary, validation is the next quarter, and testing is the following quarter. The twelve test quarters run from 2023 Q1 through 2025 Q4, yielding $T = 749$ pooled test days. Because the label is a 21-trading-day forward return, we purge the final 21 training days to eliminate label overlap [17]. A secondary sliding 252-day specification is only a robustness exercise and opens no additional inferential family.

3.2 Information coefficient and portfolio

Daily IC (§1) is computed against 21-trading-day forward returns, over stocks with non-missing features on that day. The headline IC is the pooled mean across all test days in the 12 folds. Spearman IC is invariant to monotone score transformations and standard in factor evaluation.

For the portfolio exercise, each day we buy the top decile and short the bottom decile of predicted scores, with equal-weighted dollar-neutral positions rebalanced every 21 trading days. The long-short Sharpe ratio is computed from non-overlapping 21-day rebalance returns and annualized by $\sqrt{252/21}$. Daily-IC dependence is handled separately by the block = 21 d bootstrap (§3.3).

Net Sharpe subtracts one-way transaction costs at each rebalance:

$$N = G - \tau_{L1}c/10^4,$$

where N and G are net and gross rebalance returns, c is the one-way cost in bps, and $\tau_{L1} = \sum_i |\Delta w_i|$ is the realized per-rebalance $L1$

turnover, at most 4 under a full long-short rotation and measured per arm and fold, so an arm that rotates positions more pays a proportionally larger cost. We report $c = 0, 5, 10, 15, 20$, and 30 bps; the headline cost is 10 bps. Net Sharpe is a descriptive economic-sensitivity layer, computed from the same tuned 12-fold confirmatory cells rather than from an earlier pilot.

3.3 Two confirmatory families

The two confirmatory families are kept statistically separate: Family-1 asks whether any tuned model beats tuned LightGBM after model-selection adjustment; Family-2 asks whether changing only the edge set changes IC at a fixed operating point, since a tuned model comparison need not identify the edge’s own contribution.

Family-1 (predictive). The benchmark is the tuned LightGBM rung (L0), and the candidate set contains the other nine tuned rungs. We apply Hansen’s SPA test using the consistent variant, loss = −daily IC, and a stationary bootstrap with block length 21 trading days.

We also report local paired evidence from DM/HLN tests on 20 pre-registered contrasts: five ladder contrasts, $\{L1-L0, L2-L1, L6-L2, L7-L2, L2s-L2\}$, and five edge-set contrasts, $\{L3-L2, L4-L2, L5-L2, L5-L4, L5-L3\}$, each in both universes. The main correction is BH-FDR at $q = 0.05$ over the pooled 20-test family. A per-universe BH correction gives the same rejection decisions.

The DM statistic uses Newey–West HAC standard errors [18] with the Newey–West (1994) data-driven bandwidth $L = \lfloor 4(T/100)^{2/9} \rfloor$ ($L = 6$ at $T = 749$) to absorb serial correlation in the daily loss differential. The HLN factor, $\sqrt{(T+1-2h+h(h-1)/T)/T}$ with $h = 21$, is a separate small-sample t -correction for the h -step forecast horizon, not a second autocorrelation adjustment. The two operate on different margins, the HAC long-run variance and the small-sample degrees of freedom, and are therefore not double-counted; the small-sample factor is conservative.

Family-2 (edge attribution). A tuned-vs-tuned edge comparison does not by itself identify the contribution of the edge, because the compared models may differ in both graph structure and effective capacity. We therefore add FC arms, the experiment label for the fixed-operating-point arms, that run the L3, L4, and L5 edge sets at L2’s frozen tuned hyperparameter vector, changing only the edge set. The matched ΔIC is an edge effect at this L2 operating point.

One caveat: although we keep the FC experiment label, the intervention fixes an operating point rather than model capacity literally. A denser edge set may be underfit under L2’s hyperparameters, so part of the 0/6 non-significance could reflect fixed-operating-point underfitting rather than a genuinely null edge effect. Inference uses paired, fold-level, seed-averaged ΔIC over $n \approx 12$ fold blocks, with 5000 stationary block-bootstrap replications and BH-FDR control over six two-sided fold-block tests. The per-row confidence intervals are unadjusted and descriptive.

Power. For each contrast, we report a minimum detectable effect (MDE) to distinguish statistical non-rejection from evidence of a precisely estimated zero. We define $MDE = 2.8 \times SE_{\text{block}}$, with $n_{\text{eff}} = T_{\text{days}}/21$. The multiplier $2.8 \approx z_{0.975} + z_{0.80}$ is the standard two-sided 5% test, 80% power factor for a z -test under an

approximately normal block-bootstrap sampling distribution. This MDE rests on a coarse non-overlapping block SE over $n_{\text{eff}} \approx 36$ blocks and is a deliberately conservative power yardstick, not the HAC daily-loss SE behind the DM/HLN p -values; “underpowered” means the gap is small relative to block-level dispersion, not that the daily DM test lacks resolution.

Three sample-size notions enter the analysis: daily-IC inference uses $T = 749$ pooled test days, the IC/Sharpe MDE uses $n_{\text{eff}} = T/21 \approx 36$ non-overlapping blocks, and Family-2 uses about 12 fold blocks.

Inference unit. Before testing, both families average the ten seeds into one series: per-day for Family-1 and per-fold for Family-2. The unit of inference is therefore the day in Family-1 and the fold in Family-2. This matches the pre-registered seed-mean estimand but does not propagate the cross-seed dispersion discussed in Section 1; that dispersion is reported separately as a single-seed deployment caveat.

3.4 Pre-registration

The protocol and tuned hyperparameters were frozen before the confirmatory run, with the hyperparameter file identified by md5 59ddd0a2. Only the two families described above are used for confirmatory inference. Earlier analyses, including horizon ablations, the loss-function horse race, and the Plan-AAA factor ranking, are disclosed as exploratory and do not enter the confirmatory decision rules. Net Sharpe is reported as a descriptive economic layer; IC is the only confirmatory metric, and we do not define a separate BH-FDR family for Sharpe.

4 DATA AND SETUP

The stock universe is a fixed survivor snapshot of the S&P 500: the 501 names (of the 503 share-class tickers listed at the snapshot date) with continuous price, sector, and news histories from 2021 through 2026. It is not point-in-time membership. Firms that left the index are absent rather than retained with delisting returns, and missing forward-return cells are excluded as NaN rather than zero-filled. The estimand is therefore this fixed survivor universe, not the point-in-time S&P 500 (Limitation L8).

The data frequency is daily close-to-close. The label, fixed across all experiments, is the cross-sectionally z -scored, market-excess transform of the 21-trading-day forward return

$$r_{T \rightarrow T+21} = p_{T+21}/p_T - 1,$$

i.e., $r_{T \rightarrow T+21}$ demeaned by the day’s cross-sectional mean and then z -scored. Predictions for day T use features measured strictly at $T-1$, leaving a one-trading-day execution lag between the last feature timestamp and the beginning of the return window.

We use the same ten canonical seeds in all experiments. Each ladder rung changes one nominal design axis relative to the rung below it (Table 1), and each rung is tuned independently with 30 Optuna trials. Equal budget equalizes trial count, not search adequacy. Neural arms have higher-dimensional spaces than LightGBM, a caveat for the “MLP beats GAT” interpretation. A trials-sensitivity sweep probes it: re-tuning L2 at $3 \times$ the budget (90 trials, matching the MLP’s search density) leaves the leak-free Universe-B penalty significant ($L2-L1 = -0.0090$, HLN $p = 0.002$, BH-reject) but drops Universe-C’s below significance (-0.0069 , $p = 0.059$,

Table 1: The tuned model ladder. Each rung changes one nominal design axis; every arm is independently retuned, so rung contrasts compare tuned operating points, not capacity-matched pairs (§3.3).

Rung	Model / graph	Targets
L0	LightGBM (tuned), no graph	non-neural benchmark
L1	MLP, no graph	value of going neural
L2	GAT + $\alpha 1$ correlation	value of adding a graph
L2s	GraphSAGE-Mean + $\alpha 1$	aggregation control vs L2
L3	GAT + $\alpha 1$ Unews	marginal value of news edges
L4	GAT + $\alpha 2$ (+ sector)	marginal value of sector edges
L5	GAT + $\alpha 4$ (all edges)	edge-stacking complementarity
L5s	GraphSAGE-Mean + $\alpha 4$	aggregation control, full edges
L6	full self-attention, no mask	attention vs structure (dense)
L7	HATS-3R-adapt	relation-attention representative

sign unchanged); the gap narrows without closing, so the primary leak-free result is not an equal-budget artifact.

We construct four graph specifications. Graph $\alpha 1$ is a 126-day rolling Spearman-correlation graph, refreshed every 21 days from training data only, with an edge when $|\rho| > 0.6$. Graph $\alpha 2$ is static GICS sector connectivity. The news graph is point-in-time: two stocks share an edge on day T if they are co-mentioned in an EODHD news-feed article whose publication timestamp, rather than event date, falls before the NYSE close on $T - 1$. The news graph fits no construction parameters: ticker tags are the provider’s per-article annotations, an edge requires only co-mention in a multi-ticker article, and the only temporal parameter is the daily cutoff—nothing is estimated from full-sample or future data. Graph $\alpha 4$ stacks the correlation, sector, and news edges.

The confirmatory run consists of 2160 main model-fold-seed cells, 240 L7/HATS cells, and 720 Family-2 fixed-operating-point cells. Each output store is complete, with no failed cells and no duplicate `cell_id`, and each feeds only its pre-specified family. The model classes are GAT [24], GraphSAGE-Mean [10], MLP, and LightGBM. L7 follows an R-GCN-style relation-specific parameterization [22], and SPA is computed with arch [23].

Pipeline positive control. A pre-confirmatory planted-signal control rules out a grossly nonfunctional graph path: with synthetic node features and a label planted on the frozen correlation topology (achievable IC 0.047), GAT recovers 82% and GraphSAGE-Mean 91% of the achievable IC while a feature-matched MLP recovers ≈ 0 (BH-significant; positive in all 20 fold-seed cells). The null is therefore not a broken-pipeline artifact. The control ran on the pilot pipeline the confirmatory runners reuse import-only; it certifies gross operation, not tuned-arm optimality.

5 RESULTS

The results point in five directions, presented in order: several arms’ bootstrap CIs exclude zero, yet Hansen’s SPA test does not confirm that any tuned arm beats LightGBM after model selection; the local DM ladder places the strongest predictive gain in the MLP rather than in the correlation graph; the edge-attribution family finds no BH-FDR rejection, with all edge contrasts underpowered; and the two central Universe-C contrasts survive a 10bps cost convention, while the news-edge IC contrast does not carry to net Sharpe.

Table 2: Per-arm pooled IC with 95% block-bootstrap CI. “e0” marks CI excluding zero. The C/L5s Univ-C value (0.0018) is conditional on its *defined* cells: 27.5% of its cells collapse to a mathematically undefined IC (Section 5.6) and are excluded pointwise, not zero-filled.

Rung	Univ B IC	e0	Univ C IC	e0
L0 LightGBM	0.0228	y	0.0195	n
L1 MLP	0.0371	y	0.0343	y
L2 GAT(corr)	0.0238	n	0.0224	n
L2s SAGE	0.0317	y	0.0313	y
L3 +news	0.0089	n	0.0101	n
L4 +sector	0.0157	n	0.0386	y
L5 +both	0.0138	n	0.0375	y
L5s SAGE all	0.0272	n	0.0018	n
L6 full-attn	0.0290	y	0.0397	y
L7 HATS	0.0147	n	0.0325	n

5.1 Headline IC ladder

Figure 1 and Table 2 report seed-averaged pooled IC by arm, with 95% block-bootstrap confidence intervals over $T = 749$ test days. Several arms have intervals that exclude zero, most consistently the non-graph MLP (L1, in both universes) and the full-attention arm (L6). The two pure correlation-GAT rungs (L2) do not exclude zero in either universe. A non-zero per-arm IC is not the same as beating the benchmark (§5.2).

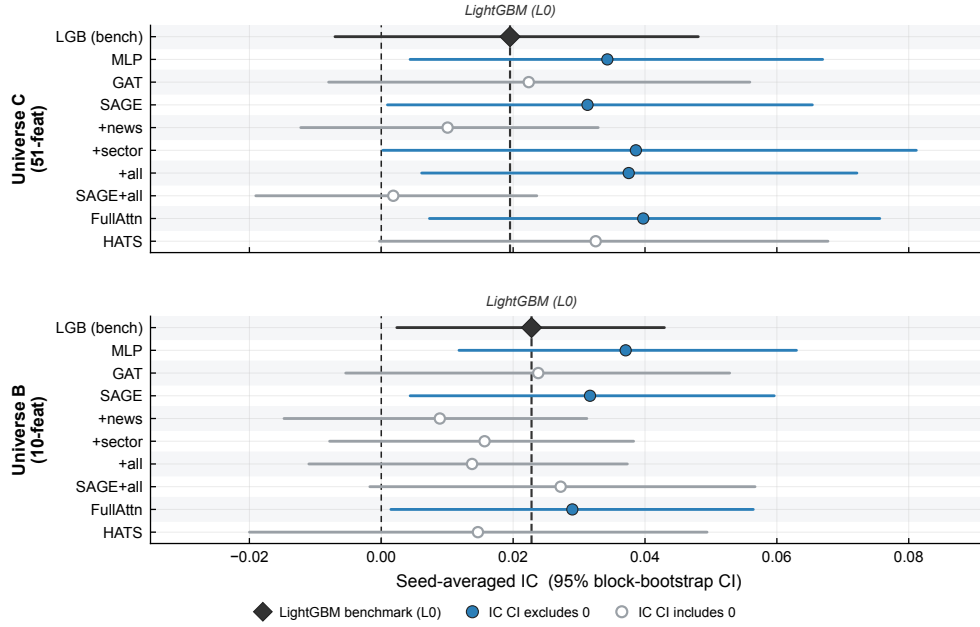
One arm requires special treatment. C/L5s has pooled IC ≈ 0.002 over its defined cells because 27.5% of its cells collapse to a mathematically undefined IC (Section 5.6). The arm remains in the $M = 9$ SPA candidate set. Undefined days are excluded pointwise rather than zero-filled, so the candidate count remains unchanged and the arm is not dropped from the family.

5.2 The predictive family: SPA and DM/HLN

Hansen’s SPA test does not reject the null that no candidate beats LightGBM in either universe ($p_{\text{consistent}} = 0.2767$ for B and 0.0774 for C; $M = 9$, $T = 749$). In Universe C the SPA bracket is $p_{\text{lower}} = 0.055$, $p_{\text{consistent}} = p_{\text{upper}} = 0.077$ —even the lower bound stays above 5%—and the comparison is underpowered (observed gaps below the 80% MDE). The selection-adjusted conclusion: no tuned arm is confirmed to outperform tuned LightGBM.

The local DM/HLN ladder gives a more detailed, but narrower, view of the same experiment (Figure 2, Table 3). We report each layer at its own error rate and do not claim a joint family-wise rate spanning both: SPA is the confirmatory answer to the global benchmark question, and the DM ladder is pre-registered local evidence at its own FDR level. Two post-hoc sensitivities (not replacements for the pre-registered families) bound the layering: pooling the 20 DM and six Family-2 p -values into one BH family rejects the same 11 DM contrasts and no Family-2 contrast, and Benjamini-Yekutieli [3], valid under arbitrary dependence, keeps 7 of the 11—L2–L1 still rejects in *both* universes, while C/L1–L0 and L3–L2 drop, consistent with their suggestive and cost-sensitive labels.

Within this tuned ladder, the strongest local evidence points to the non-graph MLP rather than to the correlation graph. In Universe C, the MLP beats LightGBM locally (L1–L0 = +0.0148, BH-reject), while the correlation-GAT loses to the MLP (L2–L1 = −0.0119, BH-reject). The tuned news-edge arm also underperforms the correlation-GAT (L3–L2 = −0.0123, BH-reject). These are local contrasts, not a global claim that graphs hurt: the sector, full-attention, HATS, and combined-edge arms all recover above the

Family-1 headline: tuned IC is small and overlaps across all arms (incl. LightGBM)

Levels only; the formal cherry-pick-robust test (no arm beats L0) is the SPA/DM figure.

Figure 1: Per-arm pooled IC for the L0-L7 ladder. Intervals are 95% stationary block-bootstrap CIs (block = 21 d, 5000 reps); seeds = 10, folds = 12, benchmark = L0.

correlation-GAT in Universe C, and the SPA non-rejection remains the selection-adjusted summary. As above, the positive Universe-C MLP-LightGBM contrast is only suggestive because the feature basis was leak-selected (Limitation L1). The negative graph contrasts are less exposed to that concern, because selection leakage can inflate apparent predictability but does not explain why the graph underperforms the MLP; consistent with this, the L2-L1 penalty is BH-significant in the clean price-volume Universe B as well ($\Delta\text{IC} = -0.0133$). Power is also asymmetric across the two directions: the correlation-graph penalty CL2-L1 exceeds its 80% MDE, a detectable effect, whereas the MLP's gain over LightGBM CL1-L0 falls below its own MDE. The data therefore identify a detectable graph *penalty* at this operating point while leaving any benchmark-beating *benefit* too small to detect at this sample size. Under-search of the higher-dimensional graph arms at the fixed 30-trial budget (§4) is an alternative reading of the negative contrasts, bounded by two observations: the same-budget, same-space GAT arms L4 and L5 land *above* L2 in Universe C (+0.0163, +0.0152, both BH-reject), and the planted-signal control shows the stack converts graph-borne signal when present.

5.3 Regime concentration

Pooled IC masks the episodic nature of cross-sectional predictability. Most positive signal is concentrated in a few quarters, especially 2024 Q4 and 2025 Q2; roughly half of the test quarters are near zero or negative for almost every arm. The pooled estimate is therefore an average over a nonstationary sequence. Single-quarter Sharpe ratios are even more fragile because they use few non-overlapping rebalance returns. This concentration

Table 3: DM/HLN pairwise ladder: seed-averaged daily ΔIC and HLN p -value (* = BH-FDR reject at $q = 0.05$ over the 20-test family). Near-equal ΔIC can differ in p as the daily-loss variance differs by universe.

Pair (A-B)	B ΔIC	B p	C ΔIC	C p
L1-L0	+0.0143	0.052	+0.0148	0.011*
L2-L1	-0.0133	< .001*	-0.0119	< .001*
L6-L2	+0.0052	0.356	+0.0174	0.001*
L7-L2	-0.0091	0.195	+0.0102	0.002*
L2s-L2	+0.0079	0.118	+0.0089	< .001*
L3-L2	-0.0149	0.006*	-0.0123	0.009*
L4-L2	-0.0081	0.227	+0.0163	0.012*
L5-L2	-0.0100	0.110	+0.0152	< .001*
L5-L4	-0.0019	0.172	-0.0011	0.831
L5-L3	+0.0049	0.066	+0.0275	< .001*

affects the pooled IC *levels* but not the load-bearing pairwise contrasts: in a leave-one-quarter-out check, none of the four BH-rejected ladder contrasts (L1-L0, L2-L1, L3-L2, L5-L3) changes sign when any single fold is removed (0/12 in both universes), so the graph-underperformance result is not an artifact of the 2024 Q4 or 2025 Q2 regimes. A leave-one-seed-out check mirrors this along the seed axis: no BH-rejected contrast changes sign when any single seed is removed (0/10 each); per-seed signs agree in 8/10 for the two Universe-B penalties, 9/10 for their Universe-C counterparts, and 10/10 for CL1-L0 and CL5-L3.

5.4 Net-of-cost crosswalk

We re-express each pre-registered ranking contrast under a 10 bps net-of-cost convention (Table 4, Figure 3). IC remains the only confirmatory metric; the net-Sharpe exercise is descriptive. The two load-bearing Universe-C ladder contrasts (CL1-L0 and CL2-L1)

Family-1: no tuned arm is confirmed to beat tuned LightGBM

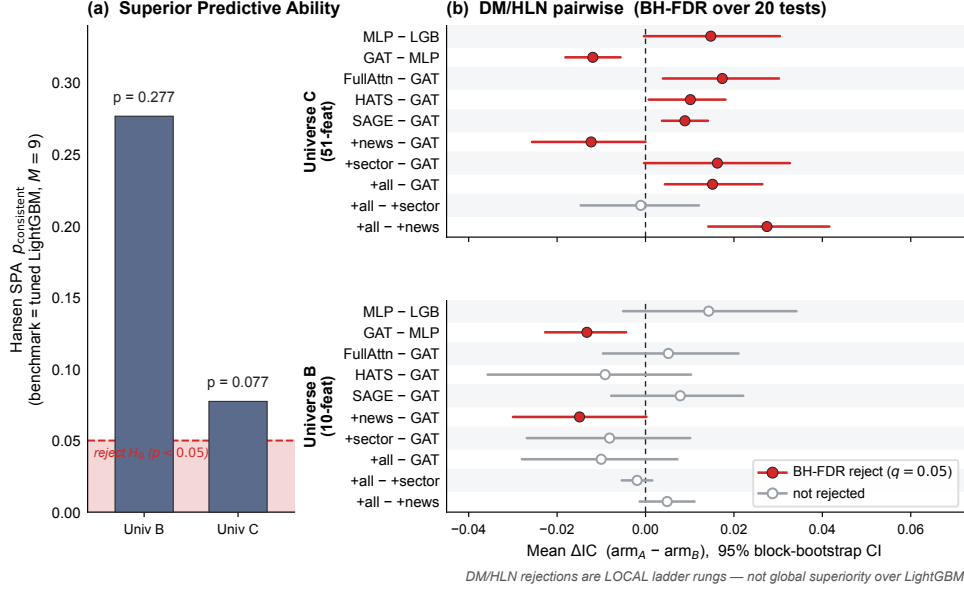


Figure 2: SPA and local DM/HLN evidence. Left: SPA $p_{\text{consistent}}$ by universe. Right: 20 pre-registered DM/HLN contrasts, with filled markers denoting BH-FDR rejection at $q = 0.05$.

survive this economic layer, with paired net ΔSharpe confidence intervals excluding zero; the edge-stacking contrast C L5–L3 also carries (a tuned-ladder pair, so edge set and operating point change together), and only the news-edge contrast C L3–L2 does not (Table 4).

For L1–L0, the economic separation is larger than the IC gap alone suggests. At 10bps, from ≈ 36 non-overlapping rebalance returns, tuned LightGBM has net Sharpe -0.22 (95% CI $[-0.82, +0.36]$), whereas the MLP has $+0.95$ ($[+0.18, +1.77]$). Because both level intervals are wide and overlap, the load-bearing object is the *paired* per-fold net ΔSharpe (Table 4): its point estimate equals the difference of the level means ($+1.17$), but its CI $[+0.36, +2.08]$ is a stationary block-bootstrap interval over the 12 paired fold differences rather than a difference of the two level intervals, and excludes zero. It declines mildly with cost because the MLP turns over more (realized L1 turnover 2.90 versus 2.25 per rebalance in Universe C).

The exception is C L3–L2. The news-edge rung underperforms the correlation-GAT in IC, but not in net Sharpe: net ΔSharpe is $+0.08$ with CI $[-0.77, +0.89]$. We read this ranking-level underperformance as economically fragile—a local IC result, not a causal or economic statement about news edges (§5.5). The crosswalk covers only Family-1 ranking pairs; Family-2 fixed-operating-point contrasts are evaluated only in IC.

5.5 The edge-attribution family: fixed-operating-point edges

Family-1’s news-edge result mixes the edge change with changes in the tuned model. Family-2 evaluates each edge set at the frozen

Table 4: Gross/net crosswalk for the two load-bearing Universe-C ladder contrasts (C L1–L0, C L2–L1) and the edge contrast C L5–L3. “Carries?” asks whether the gross-IC result reproduces in net Sharpe; “CS” (cost-sensitive) marks a gross-IC contrast that does not carry because its net ΔSharpe CI straddles zero.

Pair	Gross ΔIC (BH)	Net $\Delta\text{Sharpe}@10\text{bps}$ [CI]	Carries?
C L1–L0	$+0.0148$ (r)	$+1.17$ $[+0.36, +2.08]$	yes
C L2–L1	-0.0119 (r)	-0.72 $[-1.42, -0.10]$	yes
C L3–L2	-0.0123 (r)	$+0.08$ $[-0.77, +0.89]$	no (CS)
C L5–L3	$+0.0275$ (r)	$+0.88$ $[+0.12, +1.65]$	yes

L2 hyperparameter vector (Table 5, Figure 4). None of the six contrasts survives BH-FDR, and all six are underpowered ($|\text{matched } \Delta\text{IC}| < \text{MDE}@80\%$). The point estimates are directionally positive, but not statistically reliable.

The sign reversals in Universe B are suggestive but not conclusive: matched ΔIC is positive where tuned ΔIC was negative. In Universe C, the sector ($+0.0137$) and sector+news ($+0.0141$) contrasts have the same sign as their tuned counterparts, while the news contrast is small and positive ($+0.0011$) despite a negative tuned rung (-0.0123). Thus, the fixed-operating-point test does not reproduce the tuned ladder’s apparent news-edge harm. Given low power, this is a failure to detect harm, not evidence that the edge is harmless.

5.6 A tuned-configuration stability failure

The Universe-C SAGE-Mean control arm with all edges (C/L5s) degenerates to constant predictions in 27.5% of its test fold-seeds: 25 fully collapsed and 8 partially collapsed cells out of 120. This is not

After costs (net @10bps): headline claims hold; among BH-significant claims only "news hurts ranking" (C) reverses (gross IC harm $\rightarrow$ net = 0)

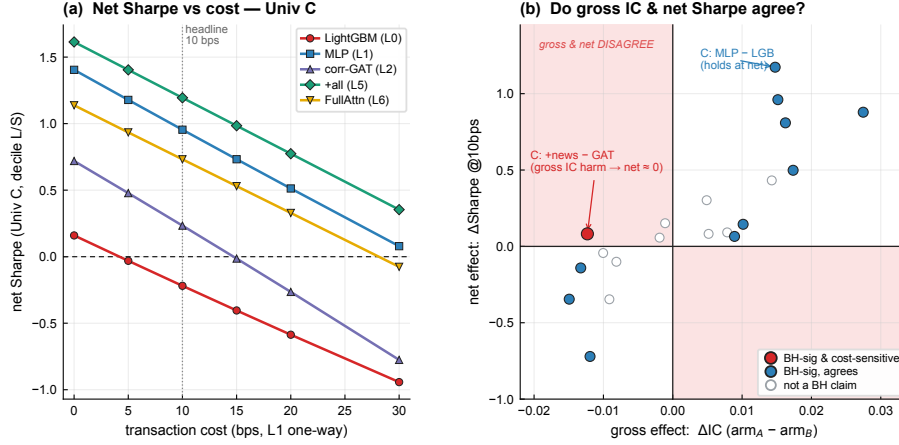


Figure 3: Net-of-cost crosswalk. Left: Universe-C cost ladder over 0–30 bps. Right: sign agreement between gross IC and net Sharpe across the 20 pre-registered pairs, with C L3–L2 annotated.

Table 5: Family-2 fixed-operating-point edge contrasts (matched Δ IC, BH-FDR over 6). UP = underpowered.

Univ	Edge	Matched Δ IC	BH	UP	Tuned Δ IC
B	news	+0.0015	n	y	-0.0149
B	sector	+0.0055	n	y	-0.0081
B	sector+news	+0.0042	n	y	-0.0100
C	news	+0.0011	n	y	-0.0123
C	sector	+0.0137	n	y	+0.0163
C	sector+news	+0.0141	n	y	+0.0152

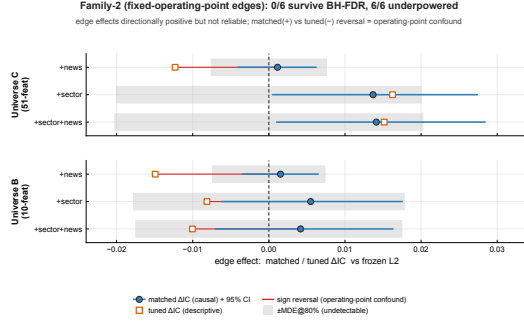


Figure 4: Matched- Δ IC versus tuned- Δ IC for the six edge contrasts. The $\pm$ MDE@80% band contains every matched estimate, and no contrast survives BH-FDR.

a training crash: affected cells report convergence but plateau at no signal, leaving Spearman IC undefined.

A plausible but unproven mechanism is oversmoothing: SAGE-Mean aggregation over a dense edge set, with tuned dropout 0.5, can smooth away cross-sectional variation; the failure is observed for SAGE-Mean with all edges, whereas L2 is a GAT on the sparse correlation graph.

We treat undefined IC as missing rather than as a measured zero, and we do not retune only this losing arm. The main conclusion is unchanged: under exclusion, zero-fill, or zero-skill handling, C/L5s mean IC remains approximately zero and the Universe-C SPA p -value remains between 0.077 and 0.080.

5.7 Exploratory failure modes

The analyses in this subsection are outside the confirmatory families. In an earlier loss-function horse race, where MSE was later fixed for the confirmatory ladder, ListMLE showed a regime-shift inversion: per-cell mean IC was -0.0458 , compared with $+0.0113$ for MSE. A plausible explanation is softmax likelihood overweighting the in-sample rank order, which can invert under regime shift.

A separate issue is Universe-C’s feature-basis leakage: only 5 of its 15 Plan-AAA Alpha158 groups survive strict $T - 1$ re-ranking, so its positive results (MLP > LightGBM, positive edge effects) are suggestive pending leak-free selection, while null and negative contrasts are less exposed (detailed in Limitation L1).

6 DISCUSSION

A per-arm bootstrap CI and Hansen’s SPA test answer different questions: one evaluates a point estimate against zero, the other whether the best candidate beats a benchmark after selection adjustment. The evidence does not show that GNNs fail. It shows that, over this sample and protocol—with the graph path itself checked by a planted-signal control (§4)—no tuned arm is confirmed to outperform a tuned tree benchmark, and the strongest local neural gain comes from the MLP.

The cleanest local DM contrast in both universes is L2–L1 < 0: adding a correlation-GAT to the same node features reduces IC relative to the MLP; a $3\times$ -budget trials sweep (§4) narrows but does not close this gap in the leak-free universe, so it is not merely an equal-budget artifact, even if exhaustive tuning fairness is not established. The C/L5s collapse offers one possible mechanism: dense-edge aggregation plus high dropout may smooth away cross-sectional signal. This mechanism is only suggestive, since the collapse is observed for SAGE-Mean with all edges and not directly for L2’s sparse-correlation GAT. The claim is therefore bounded to the correlation graph at the tuned operating point, not graph architectures in general.

The news-edge result shows why the two-family design matters. In Family-1, the tuned news-edge rung appears to harm ranking. In Family-2, the same edge change at the fixed L2 operating

point is tiny, statistically insignificant, and in Universe B oppositely signed. The tuned ladder answers a deployment question; the fixed-operating-point arms answer an edge-attribution question, subject to underfitting. Treating them as the same would overstate an edge ablation; the net-Sharpe exercise points the same way.

7 LIMITATIONS AND FUTURE WORK

(L1) *Selection-stage leakage in Universe-C's basis.* Universe C's 51 columns come from Plan-AAA top-15 Alpha158 groups ranked under same-day OHLC. Under strict $T - 1$ correction, only 5 of the 15 groups remain in the top 15. Runtime features are measured at $T - 1$, but feature-basis selection is leaked, so Universe-C positive results, including $\text{MLP} > \text{LightGBM}$ and positive fixed-operating-point edge effects, are suggestive pending leak-free re-selection. Null and negative within-universe contrasts are less exposed because selection leakage can inflate signal but does not explain graph underperformance.

(L2) Predictive skill is concentrated in a minority of the 12 test quarters, so fold-level results are needed to interpret pooled IC. **(L3)** The C/L5s constant-prediction collapse is reported as a stability finding rather than repaired by retuning. **(L4)** The LightGBM comparison is underpowered (observed gaps below the MDE): no reliable evidence of superiority, not proven equality.

(L5) Net Sharpe is descriptive and sensitive to heavy tails and turnover; economic interpretation should rely on fold-level paired ΔSharpe . The equal-weight decile portfolio ignores liquidity constraints and short-borrow costs, so net Sharpe is an upper bound on realizable performance.

(L6) L6 represents the dense learned-attention family [5, 15]; it does not cover learned-sparse graph architectures such as FinMamba [4]. We also do not benchmark LSTM or Transformer time encoders. **(L7)** The evidence comes from one market and one prediction horizon.

(L8) *Survivorship and non-point-in-time composition.* The universe is a fixed end-window survivor snapshot, not point-in-time (PIT) membership. A public-record constituent audit, rather than official S&P DJI or CRSP history, identifies two approximate deviations. First, 87 window members left before the snapshot date and are absent, equal to 14.8% of ever-member names and 8.2% of PIT member stock-days. Because reported results are within-universe paired contrasts on the same snapshot, this reduces mechanical cross-arm imbalance but does not make the estimand PIT. Pairing shields only cross-arm *differences*: the few *level* statements (per-arm IC excluding zero, Table 2; per-arm net Sharpe, §5.4) are unshielded and inflated by the survivor-and-look-ahead universe, so we read them as within-sample descriptive, not PIT-realizable, levels. Attrition is non-uniform, and omitted names include 2023 regional-bank failures and acquisition targets. Second, the snapshot over-includes current constituents before index-addition dates, creating 8.1% look-ahead stock-days and a two-sided composition mismatch of 16.3%. Paired contrasts do not remove this sampling bias. The mismatch is bounded in liquidity terms (among 76 mid-window additions, the smallest market capitalization is \$6.6 B and none is below \$5 B), but not in distress terms: SVB, Signature, and First Republic remain omitted.

(L9) The correlation-graph hyperparameters are fixed rather than ablated ($|\rho| > 0.6$, 126-day window); a sensitivity sweep is left for future work.

The next steps are a point-in-time universe rebuild with delisting returns, a longer sample, cross-market replication, intraday rebalancing for short-window news, and learned-sparse architectures evaluated under the same two-family protocol.

8 REPRODUCIBILITY

The project repository versions the code, confirmatory outputs, frozen tuned hyperparameters, the fold calendar, the Optuna grid, the ten seeds, and all analysis and figure scripts. A numeric-claim verifier checks the draft against underlying outputs, and the pre-registered protocol fixes the two-family decision rules.

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